



Product Rule :-

$$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

$\Downarrow ?$

$$\frac{d^n}{dx^n}(uv)$$

Successive Differentiation :-

$$y_1, y_2, \dots, y_n$$

Leibnitz's Thm :-

If each of the functions $u = u(x)$ and $v = v(x)$ are differentiable n -times then,

$$(uv)_n = u_n v + nC_1 u_{n-1} v_1 + nC_2 u_{n-2} v_2 + \dots + nC_r u_{n-r} v_r + \dots + uv_n$$

$$= u_n v + n u_{n-1} v_1 + \frac{n(n-1)}{2} u_{n-2} v_2 + \dots + \frac{n!}{r!(n-r)!} u_{n-r} v_r + \dots + uv_n$$

Example:- If $y = \tan^{-1} x$ then, Show that,

$$(1+x^2)y_{n+2} + 2(n+1)y_{n+1}x + n(n+1)y_n = 0$$

Solⁿ Given,

$$y = \tan^{-1} x$$

$$\Rightarrow y_1 = \frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2)y_1 = 1 \quad \text{--- (1)}$$

Differentiating both sides of (1)

$$(1+x^2)y_2 + y_1 2x = 0$$

$$\Rightarrow \frac{(1+x^2)}{v} \frac{y_2}{u} + \frac{2x}{v} \frac{y_1}{u} = 0$$

--- (2)



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Applying Leibnitz's thm of each part of (2) $\Rightarrow$

$$\begin{aligned} & (1+x^2)y_{n+2} + n(2x)y_{n+1} \\ & + \frac{n(n-1)}{2} (2) y_n + 2x y_{n+1} \\ & + 2ny_n = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow & (1+x^2)y_{n+2} + 2nx y_{n+1} \\ & + n(n-1)y_n + 2x y_{n+1} + 2ny_n \\ & = 0 \end{aligned}$$

$$\begin{aligned} \Rightarrow & (1+x^2)y_{n+2} + (2nx + 2x)y_{n+1} \\ & + (n^2 - n + 2n)y_n = 0 \end{aligned}$$

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$$\Rightarrow (1+x^2)y_{n+2} + 2x(n+1)y_{n+1} + (n^2+n)y_n = 0$$

$$\Rightarrow (1+x^2)y_{n+2} + 2x(n+1)y_{n+1} + n(n+1)y_n = 0$$

(Proved)